

Describing Meaning — Week 8

Davidsonian and Neo-Davidsonian Event Semantics

1. Why Davidsonian event semantics?

Adverbs and adjunct PPs create a problem for a simple predicate-logic treatment.

Jones buttered the toast slowly in the kitchen.

A natural idea is that the *core verbal relation* should still include the verb and its selected arguments, while adjuncts are added outside that relation.

That fits a basic syntactic contrast:

	Arguments	Adjuncts
Selected by the verb?	yes, typically	no
Usually optional?	not always	usually
Repeatable?	restricted	often yes
Examples	subject, object	adverbs, many PPs

Davidsonian Semantics:

- keep **subjects and objects inside the verbal predicate**
- add an **event variable** to that predicate
- let **adjuncts** contribute separate predicates of the same event

For example:

$$\exists e[\text{BUTTER}(j, t, e) \wedge \text{SLOW}(e) \wedge \text{IN}(e, \text{kitchen})]$$

Worked example

Mary opened the door with a key.

A Davidsonian representation:

$$\exists e[\text{OPEN}(m, d, e) \wedge \text{WITH}(e, k)]$$

The subject and object stay inside **OPEN**, while the PP contributes an extra conjunct.

Exercise 1: Davidsonian translations

For each sentence:

- a. identify the **arguments** and the **adjuncts**
 - b. give a **Davidsonian** representation
1. Lucy poured the coffee in the kitchen at midnight.

 2. Brutus stabbed Caesar violently with a knife.
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2. Conjunction in Davidsonian event semantics

Exercise 2: Davidsonian conjunction practice

Give Davidsonian representations for the following. Also say how many events your representation commits us to.

1. Lucy opened the door and Mary closed the window.

2. Mary buttered the toast slowly and carefully.

3. John opened the door and closed the window.

4. Sam opened the door and the window.

3. Neo-Davidsonian semantics

Davidsonian semantics is attractive because it respects the syntax-inspired distinction between arguments and adjuncts, it also builds *the subject and object positions into the lexical meaning of the verb*.

Neo-Davidsonian semantics takes a different approach:

$$\exists e[\text{BUTTER}(e) \wedge \text{AGENT}(e, j) \wedge \text{THEME}(e, t) \wedge \text{SLOW}(e)]$$

Now:

- the verb contributes just an **event predicate**
- participants are related to that event by **thematic-role predicates**
- adjuncts remain ordinary predicates of events

So neo-Davidsonian semantics makes the semantics of arguments and adjuncts more parallel, even though syntax still distinguishes them.

Jones buttered the toast slowly.

Davidsonian:

$$\exists e[\text{BUTTER}(j, t, e) \wedge \text{SLOW}(e)]$$

Neo-Davidsonian:

$$\exists e[\text{BUTTER}(e) \wedge \text{AGENT}(e, j) \wedge \text{THEME}(e, t) \wedge \text{SLOW}(e)]$$

Exercise 4

For each sentence, give a Neo-Davidsonian representation.

1. Mary opened the door with a key.

2. Lucy poured the coffee in the kitchen.

3. Brutus stabbed Caesar violently.

4. Argument/adjunct alternations

Consider the pair:

- (a) Mary loaded the truck with the hay.
- (b) Mary loaded the hay onto the truck.

These differ syntactically:

- in (a), *the truck* looks like the object and *with the hay* looks like an adjunct
- in (b), *the hay* looks like the object and *onto the truck* looks like an adjunct

But intuitively they have a very similar semantic core:

There is a loading event whose agent is Mary, whose theme is the hay, and whose goal is the truck.

A Davidsonian picture

A Davidsonian analysis would naturally keep the direct object inside the verbal predicate and the PP outside:

$$\exists e[\text{LOAD}(m, tr, e) \wedge \text{WITH}(e, h)]$$

versus

$$\exists e[\text{LOAD}(m, h, e) \wedge \text{ONTO}(e, tr)]$$

This makes the two sentences look less parallel semantically than we might want.

A neo-Davidsonian picture

Neo-Davidsonian semantics gives them the same representation, which is plausibly the desired result in this case.

$$\exists e[\text{LOAD}(e) \wedge \text{AGENT}(e, m) \wedge \text{THEME}(e, h) \wedge \text{GOAL}(e, tr)]$$

5. Causative / inchoative pairs

Now consider:

- (a) Mary opened the door.
- (b) The door opened.

These seem closely related. Intuitively, (a) says there was an opening event with Mary as agent and the door as theme; (b) says there was an opening event with the door as theme.

A Davidsonian picture

A Davidsonian treatment will naturally make these look like different predicate frames:

$$\exists e[\text{OPEN}(m, d, e)]$$

and

$$\exists e[\text{OPEN}(d, e)]$$

or perhaps even two different lexical predicates.

That is not impossible, but it makes the connection less transparent.

A neo-Davidsonian picture

In neo-Davidsonian semantics we can write:

$$\exists e[\text{OPEN}(e) \wedge \text{AGENT}(e, m) \wedge \text{THEME}(e, d)]$$

and

$$\exists e[\text{OPEN}(e) \wedge \text{THEME}(e, d)]$$

Now the entailment is straightforward:

$$\exists e[\text{OPEN}(e) \wedge \text{AGENT}(e, m) \wedge \text{THEME}(e, d)] \models \exists e[\text{OPEN}(e) \wedge \text{THEME}(e, d)]$$

6. Passives

Now compare:

- (a) Mary opened the door.
- (b) The door was opened.
- (c) The door was opened deliberately.

Passives are like inchoatives in that the theme appears as the surface subject. But unlike inchoatives, passives still seem to involve an agent, even when that agent is not overtly expressed.

One reason to think this is the possibility of agent-oriented adverbs such as *deliberately* or *intentionally*:

The door was opened deliberately.

This seems to imply not just that there was an opening event involving the door, but that someone carried it out deliberately.

A Davidsonian picture

This fits very naturally with a Davidsonian analysis. The active keeps the agent argument overt:

$$\exists e[\text{OPEN}(m, d, e)]$$

The passive can then be understood as existentially binding that argument:

$$\exists x \exists e[\text{OPEN}(x, d, e)]$$

So the passive does not remove the agent from the semantics altogether; it leaves us with an existentially quantified agent.

A neo-Davidsonian picture

A first neo-Davidsonian thought might be to treat the passive simply as:

$$\exists e[\text{OPEN}(e) \wedge \text{THEME}(e, d)]$$

But that is too weak to capture what "deliberately" tells us, namely that there IS an agent here, just not appearing on the surface.

To capture the agentive meaning, we seem to need something more like:

$$\exists x \exists e[\text{OPEN}(e) \wedge \text{AGENT}(e, x) \wedge \text{THEME}(e, d)]$$

Which is possible but runs counter to one of the central motivations of Neo-davidsonianism, that arguments are all introduced explicitly in the syntax.

7. Argument valence

Neo-Davidsonian semantics is attractive partly because it can make semantic participant structure more uniform.

But this does *not* mean syntax stops mattering.

Compare:

- (a) John ate.
- (b) John ate the fish.
- (c) John dined.
- (d) *John dined the fish.
- (e) *John devoured.
- (f) John devoured the fish.